

SSH Packet Tracer .PKA Lab Creation Guide

This guide explains how to create an SSH practical lab in Cisco Packet Tracer

Step 1: Create Topology

Add:

- PC
- Switch
- Router

Connect:

PC ---- Switch ---- Router

Step 2: Configure Initial IP Addressing to Client PC

Client PC Configuration:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Step 3: Router configure SSH completely:

In Router

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
! Secure privileged mode
```

```
enable secret class123
```

```
! Configure interface
```

```
interface g0/0
```

```
ip address 192.168.1.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

! Configure domain name

ip domain-name snit.local

! Generate RSA keys

crypto key generate rsa

1024

! Create SSH user

username admin password cisco

! Console password

line console 0

password admin123

login

exit

! SSH remote access

line vty 0 4

login local

transport input ssh

exit

! Enable SSH Version 2

ip ssh version 2

! Encrypt all plain-text passwords

service password-encryption

end

write

Step 4: Test SSH Connectivity

From PC Command Prompt:

ssh -l admin 192.168.1.1

Password:

cisco

Step 5: Test SSH In Router Show SSH Version

On router:

show ip ssh

Expected Output:

SSH Enabled - version 2.0

Authentication timeout: 120 secs

Authentication retries: 3

Now:

- SSH Version 2 is secure
- Uses encryption
- Safer remote administration

Expected Result

Students should Connect SSH

Step 6:After student logs in through SSH:

After student logs in through SSH:

On Client PC:

show ssh

Expected Output Example:

Connection	Version	Mode	Encryption	Hmac	State	Username
0	2.0	IN	aes128	hmac-sha1	Session Started	admin

This proves:

- SSH session established
- Encryption is active
- Authenticated user connected